SURJECTIVE INTEGRAL MAPS OVER AN OPEN EXTEND TO THE WHOLE BASE BY NORMALIZATION PLUS BOUNDARY

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 13685-13691 of sources/stacks-more-morphisms/pinned-original-source.tex; proof span is lines 13693-13702, the complete author proof (8 body lines).

Editorial note. Navigation stubs: 3 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Integral sibling of L16126 (same wave): no quasi-compactness is needed on the integral side, matching the weaker hypotheses; the author's slightly compressed phrasing (normalization of S in U) is preserved verbatim by span assembly.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. Morphisms-chapter construction of normalization (section normalization-X-in-Y) and its two structural lemmas (localization compatibility and identification over integral extensions); integrality of coproducts.

Source reference. Original lemma 03GP: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 13702–13702. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 035K: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 13702–13702. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0BAK: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 13702–13702. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let S be a scheme. If $U \subset S$ is open and $V \to U$ is a surjective integral morphism, then there exists a surjective integral morphism $\overline{V} \to S$ with $\overline{V} \times_S U$ isomorphic to V as schemes over U.*

Proof. Let $V' \to S$ be the normalization of S in U, see Morphisms, Section 0BAK. By construction $V' \to S$ is integral. By Morphisms, Lemmas 035K and 03GP we

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see that the inverse image of U in V' is V. Let Z be the reduced induced scheme structure on $S \setminus U$. Then $\overline{V} = V' \amalg Z$ works. $\square$